Week 16
Model-Based RL: Dreamer, MuZero, and Latent World Models

STUDY-RL — a self-paced Deep RL curriculum

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Learning objectives

Study companion. Read `teaching_guide.md` in this folder for step-by-step examples, an analogy with its limits, and self-checks. Then use `workbook.ipynb` to implement and vary the calculations.

Key Idea

After this week you will:

1. Distinguish PETS-style "ensemble dynamics + MPC" from latent world models like Dreamer.
2. Derive the Dreamer ELBO: reconstruction + reward + continue + KL between posterior and prior over the stochastic latent.
3. Derive the actor-critic-in-imagination update with λ-returns.
4. Implement a tiny RSSM and train an actor in imagination on CartPole; visualise imagined vs. real rollouts.
5. Summarise MuZero / EfficientZero / TD-MPC2 / DIAMOND and place them on the planning–learning axis.
6. Articulate where a learned world model is safer (and where it is not) than a model-free policy.

1 Motivation & intuition

A world model is a learned simulator. Once you have one, you can generate as many synthetic experiences as you want — this is sample-efficiency by hallucination. The model-based family ranges from PETS (probabilistic ensembles + MPC) through Dreamer (latent world model with imagination-trained actor) to MuZero (tree search on a learned latent).

Hallucination here is desirable. We feed it back to the learner; the world doesn't have to consent.

Intuition. Model-free RL is like learning to drive by trial and error on real streets. Model-based RL is building a driving simulator from your experience and then practising any manoeuvre you like inside the simulator. Dreamer is special because the simulator is *abstract* — it does not need to look like the world, only act like it under reward.

2 Mathematical formulation

2.1 The Recurrent State-Space Model (RSSM)

Definition 2.1.1 (RSSM)**.** At each step t, the agent maintains:

- a deterministic recurrent state $h_t = \mathrm{GRU}(h_{t-1}, [s_{t-1}, a_{t-1}])$,
- a stochastic latent $s_t \sim p(s_t \mid h_t)$ called the *prior*,
- posterior $s_t \sim q(s_t \mid h_t, o_t)$ when an observation o_t is available.

The model emits:

$$o_t \sim p(o_t \mid h_t, s_t), \quad r_t \sim p(r_t \mid h_t, s_t), \quad c_t \sim p(c_t \mid h_t, s_t),$$

where $c_t \in \{0, 1\}$ is the continue flag.

2.2 Dreamer world-model ELBO

Theorem 2.2.1 (Dreamer world-model objective)**.** *Index reward and continuation by arrival: action a_{t-1} leads to (o_t, r_t, c_t). For $\tau = (o_{1:T}, a_{0:T-1}, r_{1:T}, c_{1:T})$, conditioning on the actions, the negative ELBO at $\beta = 1$ is*

$$\begin{aligned} -\mathcal{L}(\theta, \phi) = \sum_{t=1}^{T} \mathbb{E}_{q_\phi} \Big[& \underbrace{-\log p_\theta(o_t \mid h_t, s_t)}_{\textit{reconstruction}} - \log p_\theta(r_t \mid h_t, s_t) \\ & - \log p_\theta(c_t \mid h_t, s_t) \Big] \\ & + \beta \sum_{t=1}^{T} \mathbb{E}_{q_\phi} \big[\mathrm{KL}(q_\phi(s_t \mid h_t, o_t) \,\|\, p_\theta(s_t \mid h_t)) \big]. \end{aligned}$$

Derivation. Start from $\log p_\theta(o_{1:T}, r_{1:T}, c_{1:T} \mid a_{0:T-1}) \geq \mathbb{E}_{q_\phi}[\log p_\theta(o, r, c, s) - \log q_\phi(s)]$, factor the joint by the temporal Markov structure of the RSSM, and gather the $-\mathrm{KL}$ terms per timestep. Choosing $\beta \neq 1$ reweights this objective and is no longer the standard ELBO identity; Dreamer V2 introduces "KL balancing" — stop-gradients on either side — to control whose distribution moves. □

Common pitfall

KL balancing controls how strongly the prior and posterior are updated; it is not a universal cure or a mathematical requirement for every RSSM. Posterior collapse can arise from the interaction of KL pressure and the decoder. The project uses free-nats clipping; exercise 5 demonstrates balancing separately.

2.3 Actor-critic in imagination

Definition 2.3.1 (Imagination rollout)**.** Given a latent state (h_0, s_0) obtained from posterior inference on real data, the *imagined trajectory* is generated by sampling $a_t \sim \pi_\psi(\cdot \mid h_t, s_t)$, stepping the prior $s_{t+1} \sim p_\theta(\cdot \mid h_{t+1})$ with $h_{t+1} = \mathrm{GRU}(h_t, [s_t, a_t])$, and reading off r_t, c_t from the model heads.

Theorem 2.3.2 (λ-return for imagined rollouts)**.** *The bootstrap target for the critic is the lambda-return*

$$G_t^\lambda = r_t + \gamma c_t\big[(1-\lambda)V_\xi(h_{t+1}, s_{t+1}) + \lambda G_{t+1}^\lambda\big], \qquad G_H^\lambda = V_\xi(h_H, s_H).$$

Actor and critic losses:

$$\mathcal{L}_\psi = -\mathbb{E}\left[\sum_t \log \pi_\psi(a_t \mid h_t, s_t) \cdot \big(G_t^\lambda - V_\xi(h_t, s_t)\big)\right], \quad \mathcal{L}_\xi = \mathbb{E}\left[\sum_t (V_\xi(h_t, s_t) - G_t^\lambda)^2\right].$$

The advantage is the lambda-return target minus the current-state value baseline. In the REINFORCE loss, stop gradients through that advantage; the critic targets are detached too. Dreamer V2/V3 replace the REINFORCE actor by a path-derivative one when actions are continuous; for discrete actions we use straight-through with Gumbel-softmax or REINFORCE as above.

2.4 PETS and MBPO — the alternative

PETS fits an ensemble of probabilistic dynamics models and plans by sampling trajectories under the model (CEM in action space). **MBPO** interleaves short-horizon model rollouts with real data in an SAC-style buffer: $D_{\text{train}} = D_{\text{real}} \cup D_{\text{model}}$. The MBPO theorem bounds the policy-performance gap using assumptions on rewards, model error, policy shift, and rollout horizon. State-distribution errors can accumulate and affect all later rewards, so total return error is not generally just one-step error times horizon. Short rollouts branched from real data reduce this accumulation.

2.5 MuZero, EfficientZero V2, TD-MPC2

MuZero learns three networks — representation h, dynamics g, prediction f (value + policy) — jointly via MCTS-derived training targets. Crucially the model is *not* trained to reconstruct observations; it is trained so the values and policies at any imagined state match the MCTS roots and chosen actions. EfficientZero V2 adds data augmentation and a self-supervised consistency loss, achieving DQN-level Atari with $\sim$2 hours of game time. TD-MPC2 combines a learned latent model with MPC at deployment and TD-learning of a value function, scaling to large multi-task settings.

2.6 DIAMOND — diffusion world models

DIAMOND (Alonso 2024) replaces the deterministic decoder with a denoising diffusion model over pixel frames, conditioned on a small encoder feature. The denoiser is the world model; the policy is trained from the denoised rollouts as in Dreamer. The trade-off: very high sample fidelity, very high compute cost.

3 Algorithms

Algorithm 1: Dreamer-lite

Input: Replay buffer $\mathcal{D}$, RSSM θ, actor ψ, critic ξ.
for *iteration = 1, ..., N* **do**
  Collect episodes with the actor (epsilon-greedy);
  for *world-model step = 1, ..., W* **do**
    Sample chunks; posterior-roll the RSSM; minimise the ELBO;
  for *behaviour step = 1, ..., B* **do**
    Posterior-roll to a batch of latent states; imagine H steps under π_ψ;
    Compute G^λ; backprop actor (REINFORCE) and critic (MSE);
  Evaluate;

Complexity: dominated by the RSSM forward pass: $O(L \cdot d^2)$ per chunk of length L and hidden width d. Imagination adds $O(H \cdot d^2)$ per behaviour step.

3.1 Worked numerical example

Take a 2-step CartPole-like rollout. Suppose the actor outputs $\pi = (0.7, 0.3)$ at h_0, s_0. It samples $a_0 = 0$, the prior produces s_1 with $r_0 = 1$, $c_0 = 0.99$. At h_1, s_1 the actor samples $a_1 = 1$ with $\log \pi = -1.2$, the prior gives $r_1 = 1$, $c_1 = 0.98$. Critic outputs $V_0 = 5, V_1 = 4, V_2 = 3$. With $\gamma = 0.99, \lambda = 0.95$,

$$G_1^\lambda = 1 + 0.99 \cdot 0.98 \cdot [0.05 \cdot 3 + 0.95 \cdot 3] = 1 + 0.9702 \cdot 3 = 3.91,$$
$$G_0^\lambda = 1 + 0.99 \cdot 0.99 \cdot [0.05 \cdot 4 + 0.95 \cdot 3.91] = 1 + 0.9801 \cdot (0.2 + 3.715) = 4.84.$$

The sampled actor loss term at $t = 0$ is $-\log \pi(a_0 = 0) \cdot (G_0^\lambda - V_0) = -\log 0.7 \cdot (4.84 - 5) = -(-0.357) \cdot (-0.16) = -0.057$ — a negative loss contribution, not the gradient itself. Differentiating with the advantage detached gives a push *away* from a_0 because the realised return was slightly worse than the baseline.

4 Implementation walkthrough

```
# World-model step: posterior roll, four-component loss.
state = rssm.initial(B, device)
for t in range(L):
    state, prior_mu, prior_std, post_mu, post_std, recon, r_pred, c_pred = \
        rssm.observe_step(state, A_oh[:, t], O[:, t+1])
    recon_loss += ((recon - O[:, t+1])**2).sum(-1).mean()
    rew_loss   += ((r_pred - R[:, t])**2).mean()
    cont_loss  += F.binary_cross_entropy(c_pred, C[:, t])
    kl = kl_gaussian(post_mu, post_std, prior_mu, prior_std)
    kl_loss    += torch.clamp(kl, min=free_nats).mean()
```

The behaviour step then opens a fresh imagination from each posterior state, computes the lambda-return, and updates actor + critic.

5 Empirical results

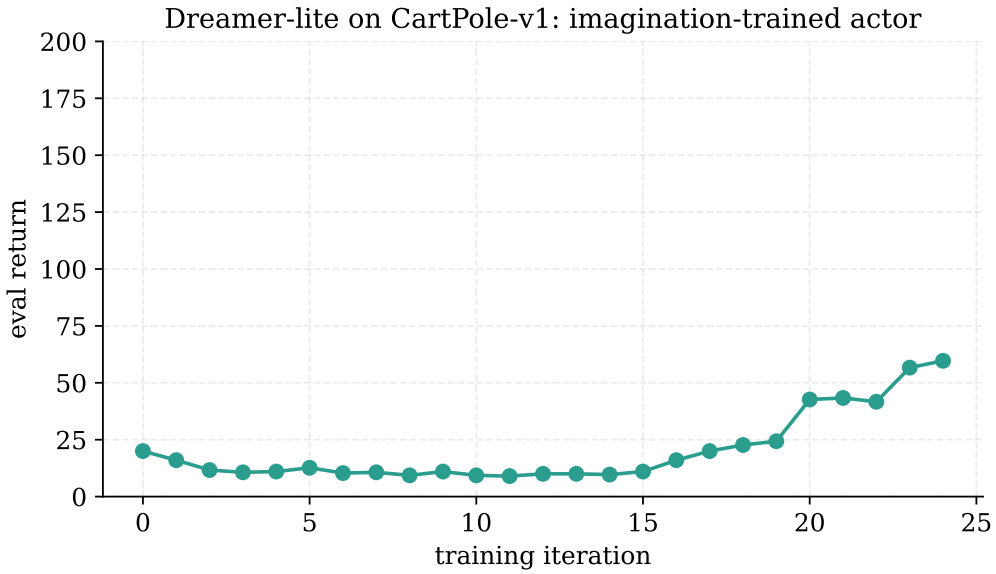


Figure 5.1: Dreamer-lite on CartPole-v1, 25 iterations ≈ 15–30s on a single GPU. Mean evaluation return ramps from ∼20 (random) to ∼60 (the cap of 200 we use to keep tests fast). Full Dreamer reaches the maximum of 500.

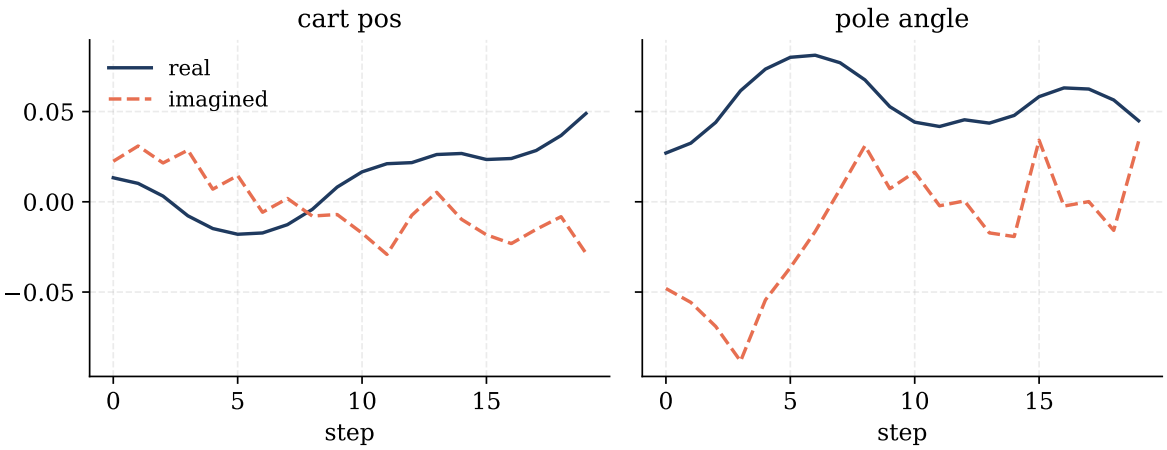


Figure 5.2: Imagined trajectory under the learned RSSM (dashed) vs. the real environment trajectory (solid) under the same actor. The world model captures the qualitative dynamics but accumulates error over time — the canonical model-based-RL tension.

Safety angle

World-model deployment failure modes.

1. **Reward hacking the model.** An imagination-trained actor will exploit any flaw in the world model: predicting infinite reward for an unreachable state, exploiting the continue head to "stay alive forever".

2. **Distributional brittleness.** The latent dynamics are accurate only on states near the training distribution. Long-horizon imagination diverges, so a policy purely improved on long imagined rollouts may collapse on real data.

3. **Mitigations.** Cap imagination horizon ($H = 15$ in Dreamer V3); use a *value-equivalence* objective (MuZero) when reconstruction is not the deployment signal; mix imagined data with a steady stream of real experience.

Citations: Hafner et al. 2023, "Mastering Diverse Domains"; Schrittwieser et al. 2020, "MuZero".

Matching application

World models for recruitment matching. The ATS analogue is a latent model of "candidate trajectory dynamics": given a stage-history of decisions, the model predicts the next state distribution under different recruiter actions. A Dreamer-style training loop would let the system simulate the consequences of hypothetical recruiter decisions before taking them — crucial for a domain where each "rollout" on a real candidate is expensive, slow, and ethically loaded. The safety angle then becomes: the world model must be *audited* for fairness in its imagined dynamics, not just its terminal outcomes.

6 Exercises

See `workbook.ipynb`.

- **Ex. 1 (⋆)** Closed-form KL between diagonal Gaussians.
- **Ex. 2 (⋆)** Lambda-return computation.
- **Ex. 3 (⋆⋆)** Build a tiny RSSM and imagine a step.
- **Ex. 4 (⋆⋆)** Run Dreamer-lite for 12 iterations.
- **Ex. 5 (⋆⋆⋆)** KL-balancing trick (Dreamer V2).
- **Ex. 6 (⋆⋆)** Compare imagined and real rollouts.
- **Ex. 7 (⋆⋆)** World-model loss components.
- **Ex. 8 (⋆⋆⋆, meaningful objective)** Recruitment world-model essay.

7 Further reading

See `extended_reading.pdf` for Chua et al. 2018 (PETS), Janner et al. 2019 (MBPO), Hafner et al. 2023 (Dreamer V3), Schrittwieser et al. 2020 (MuZero), and Alonso et al. 2024 (DIAMOND). Sourced from `research/offline_modelbased_sota.md` (Track B).